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RESEARCH ARTICLE

Deep Learning Techniques for Crater Detection on Lunar Surface Images from Chandrayaan-2 Satellite

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Abstract
Lunar exploration is pivotal in establishing a human presence on the Moon, and lunar crater detection plays a major role in this pursuit. The study is divided into two key phases: the creation of a specialized annotated dataset sourced from the Optical High-Resolution Camera on the Chandrayaan-2 satellite, and the evaluation of model performance using this dataset. Employing models such as FasterRCNN, YoloV5, and YoloV1, the investigation reveals the YoloV5 model's superiority, achieving a precision of 92% and a recall of 83% for lunar crater detection. This finding constitutes a significant contribution to lunar exploration research.

Keywords Lunar crater detection · Chandrayaan-2 · Yolo · FasterRCNN

Introduction

Over the last few decades, many lunar missions have gathered and returned data successfully. For instance, numerous studies have used data from the Apollo, Clementine, Selenological and Engineering Explorer, Chandrayaan-1, Lunar Reconnaissance Orbiter (LRO)/Lunar Crater Observation and Sensing Satellite missions which acquired a variety of data (Convenevole et al., 2020; Nozette et al., 1996). The most frequent characteristics in these analyses are impact craters. Impact craters, including ones that originated 4 billion years ago, are typically well maintained on the Moon because corrosion and weathering are much weaker there than they are on Earth (Yang et al., 2020; Hartmann, 1970). These craters are particularly important for understanding the moon's evolutionary history, the mechanisms that created the lunar surface's land-forms, and early mare volcanism AMNH (2023). Additionally, these craters depict the history of massive impact on the lunar surface. Through careful analysis of crater characteristics, including their shape, dimensions, and frequency, researchers can attain valuable insights into the underlying processes that contribute to deterioration, the factors influencing impacts, etc. Shaheen et al. (2022). As a result, impact craters are crucial to lunar studies, and several techniques have been developed to find them (Jia et al., 2020).

Lunar craters range in size from small, simple ones to large, complex structures. The size distribution of lunar craters follows a power-law relationship known as the "crater size-frequency distribution." (Toyokawa et al., 2022). This relationship reveals that as the size of craters increases, the number of craters of that size decreases exponentially. In the past, crater detection has been performed through manual examination of images. However, this method is impractical when dealing with large quantities of craters, ranging from kilometers to sub-kilometers in size. Consequently, human-generated databases have been created, which are either comprehensive in terms of spatial coverage but limited to the largest craters, or comprehensive in terms of size range (Stepinski et al., 2012; Bandeira et al., 2012). This led to automation in the field of crater detection. Automatic detection techniques are more effective than manual ones

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for detecting huge areas and craters with a sub-kilometer size (Sawabe et al., 2006). Automated techniques also enable us to detect craters in real time (Chatterjee et al., 2023; Li et al., 2011). Numerous methods have been created to address this demand, using images, and even digital elevation models (DEMs) (Jia et al., 2021). Digital elevation models (DEMs) provide a three-dimensional representation of the lunar surface by capturing a craters height and depth at various locations. They are created by a combination of data sources such as satellite images, radar, etc. It is simpler to acquire a picture of the lunar surface in high resolution than a DEM with the same resolution. As a result, image-based crater detection algorithms (CDAs) are employed more commonly than DEM-based detection algorithms. The automatic detection of craters on the lunar surface has been a topic of interest in the field of planetary science. Craters provide valuable information and their detection and analysis can help to improve our understanding of the Moon and other planetary bodies. Several studies have been conducted to develop methods for detecting craters and estimating their ages using advanced algorithms and techniques. Such CDAs have been studied in Salih et al. (2017) in which he presented a method for detecting craters and estimating their ages in mare regions on the Moon. The method uses a combination of image processing and machine learning algorithms to identify craters and estimate their ages based on their morphological characteristics. The results show that this method can effectively detect craters in mare regions and provide accurate age estimates. Hashimoto and Mori (2019), proposed a method for detecting craters on the lunar surface using a deep learning approach based on grid partition. The use of a grid partition allows the method to handle large images efficiently, as each grid cell can be processed independently. This makes it possible to apply the method to high-resolution images of the lunar surface, which can contain many craters. Wu et al. (2021) presented a method for detecting craters using a convolutional neural network to learn complex patterns in the data, which can improve its ability to detect craters under different conditions. Di et al. (2014) used a boosting algorithm to detect square regions with a single crater, followed by a Hough transform to delineate the edges of the crater within the region. Salamunićcar and Loncaric (2010) used a sub-pixel interpolation algorithm for crater detection from the elevation samples of lunar data which resulted in numerous craters being successfully detected.

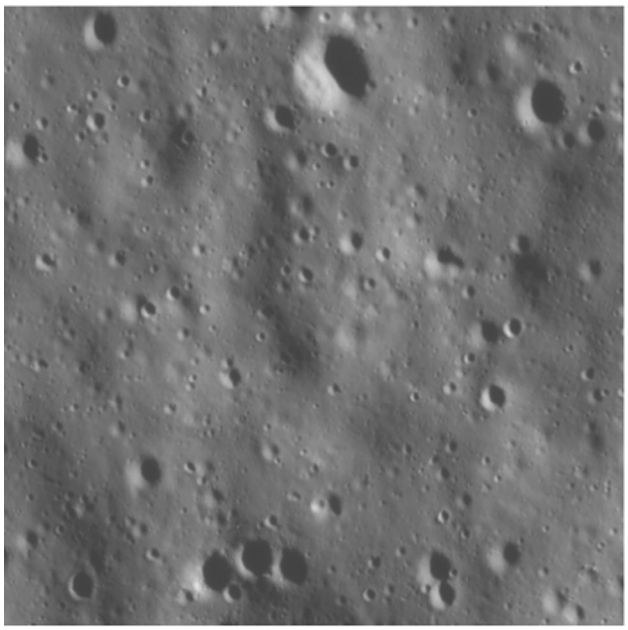


Fig. 1 Reconstructed image after patch operations. Geospatial coordinates: Top-left (4.694569, 230.985095), Lower-right (4.70014851, 230.979353)

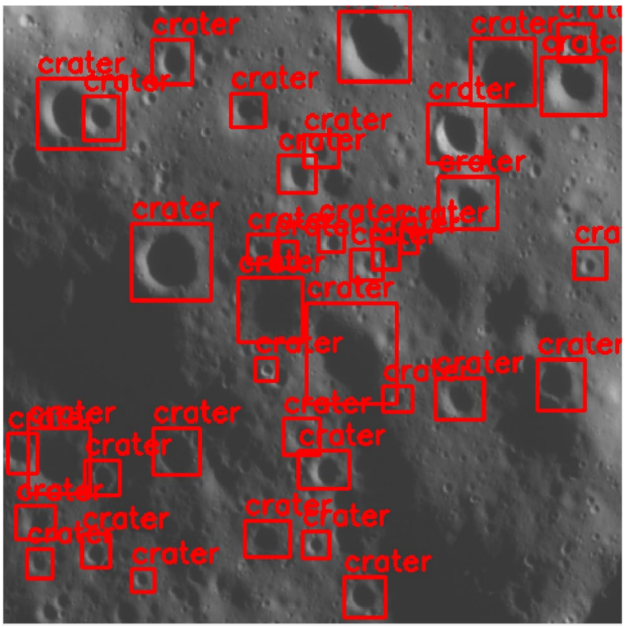


Fig. 2 Manually Labelled Image. Geospatial coordinates: Top-left (4.728046, 230.979353), Lower-right (4.733626, 230.973611)

In this work, we trained and compared multiple convolutional neural networks (CNN) to perform crater identification on high resolution lunar images. We created a custom annotated data set and trained three CNNs from scratch to identify craters on the Moon. CNNs are utilized in lunar crater detection because they excel at automatically learning and extracting relevant features from image data (Long et al., 2015). CNNs utilize their hierarchical structure and convolutional operations to recognize the unique visual attributes of craters, including their shapes, edges, and textures (Zeng et al., 2017). This capability enables the differentiation of craters from the lunar terrain around them, facilitating precise detection. The capacity of CNNs to learn and generalize from extensive datasets makes them highly

advantageous for automated crater detection, allowing for efficient and dependable analysis of expansive lunar regions.

Materials and Methods

Dataset Preparation

The raw data was acquired from the official repository of ISRO Indian Space Science Data Center (2022). The images were obtained from the Optical High Resolution Camera (OHRC) on board the Chandrayaan-2 mission satellite. The Chandrayaan-2 mission was successfully launched on 22nd July 2019 at 14:43 h by GSLV MkIII-M1 from Satish Dhawan Space Centre (SDSC), Sriharikota. OHRC is a camera that captures images in great detail in the visible light spectrum that is the visible panchromatic (PAN) band. OHRC's primary objective is to take pictures of a landing site before touchdown to help identify safe areas and avoid hazards. The coverage area of OHRC is of 12 km × 3 km with a ground resolution of 0.19 m. OHRC is an optical camera system based on Time Delay Integration (TDI) imaging sensors with 12,000 detectors. The images from the OHRC are stored in Planetary Data System version 4 (PDS4) format. A Python package called pds4-tools was used to access PDS4 data in Python. The full image obtained from the PDS4 data had a dimension of 101,074 × 12,000; smaller images of size 640 × 640 pixels were reconstructed from the full image patch by patch. The cropped images proved to have good clarity; Fig. 1 is an example of the created image. These images were resized based on the model before training.

In the geospatial domain, the analysis of images from the Chandrayaan-2 Orbiter's OHRC offers a compelling example of the complexities and capabilities in spatial data processing. The OHRC imagery used for this study, targeting the Moon, was captured on March 21, 2022, within a precise timeframe (05:25:22.587Z to 05:25:38.971Z), and includes detailed metadata crucial for geospatial analysis. The selenographic coordinates of the image's corners upper left (4.315163°, 230.985095°) and lower right (5.196323°, 230.877434°) provide a basis for mapping the image onto the lunar surface. The image was captured from an altitude of 76.66 kms, and covers an area of 43,785,256.8 m^2. The smaller images were computationally constructed from the larger ones which allowed us to calculate the coordinates of each smaller images which had a coverage of 120 m × 120 m.

Modified Open Labeling is an open-source software that was used to manually label 424 of these smaller images, which were generated and stored. These custom images were labeled in YoloV5 text format with only 1 class of objects, i.e., craters. Each crater labeled in the image stored 4 sections of data in percentage with respect to the dimension of the image. They are the x coordinate of its center, y coordinate of its center, width of the image, and height of the image.

The data from the images and labels were combined and converted into a different form for training. For object detection models, in general, the input requires the images and the bounding boxes along with the number of images in the dataset. The bounding boxes were converted to formats specified by the model used. Figure 2 shows a typical example of a labeled image.

Data Augmentation

Data augmentation is the process of artificially increasing the size of a dataset by creating new training samples through various transformations to the original data. Data augmentation is used to increase the variability and diversity of the training set Chen and Fan (2021), which can improve the performance and generalization of machine learning models. By increasing the amount and diversity of the training data, we can reduce the overfitting of the model and improve model accuracy. The data obtained after augmenting minimizes the distance between training and validation sets Shorten and Khoshgoftaar (2019) In this work, several data augmentation techniques are used to improve the robustness of their dataset for training a model to identify craters in images. These techniques include flipping, which can increase the number of samples by a factor of 8, brightness adjustment, where new samples are created by increasing or decreasing brightness by 30%, exposure adjustment, where the exposure of the images was increased or decreased by 25%, and mosaic, where multiple images were combined into a single tiled image. These techniques can help the model learn to identify craters even in images with poor lighting

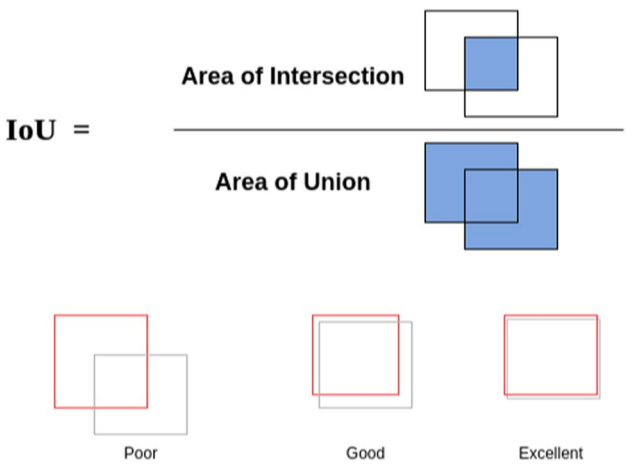


Fig. 3 IoU visualisation

or other variations. After applying these augmentations to 424 samples, a total of 852 samples were generated for the initial experiment. Further augmentations such as random resized crop which crops and resizes the image to 460 × 460 pixels, with a scale between 0.7 and 1.0 and an aspect ratio between 0.9 and 1.1, RandomBrightnessContrast which is used to randomly adjust the brightness and contrast within a limit of − 0.3 to 0.3, HueSaturationValue which alters the hue, saturation, and value within limits of 10, 20, and 10 respectively, GaussNoise which adds Gaussian noise with a variance between 10.0 and 50.0, Both MotionBlur and GaussianBlur are used which apply blur with a limit of 3–7, RandomFog which applies fog with a coefficient between 0.1 and 0.5, and finally, RandomShadow which applies a shadow in the region from the middle to the bottom of the image. These augmentations further diversify the training examples and the final dataset contained 3240 samples.

The labeled data was uploaded to roboflow. Roboflow is a cloud-based platform that provides a set of tools for managing and enhancing image datasets for machine learning. The platform included a web interface that allowed us to upload and manage our dataset. Roboflow supports a variety of data formats like JPEG, PNG, etc. and, it is capable of handling images of any size. The platform includes a range of preprocessing tools, such as resizing, normalizing, and color correction, which helped keep the custom data consistent and optimized for machine learning (ML). Once the dataset is preprocessed and augmented, Roboflow allowed us to export the data into a variety of formats, such as TensorFlow, PyTorch, and Pascal Voc, making it easy to integrate the data into a wide range of ML frameworks.

Proposed Method

The proposed study was executed employing a collection of images derived from the custom dataset. The methodology was bifurcated into two distinct phases. The initial phase involved the utilization of an initial dataset, encompassing 852 images, to evaluate three different object detection models: Faster R-CNN, YoloV1, and YoloV5. Subsequently, the second phase was conducted employing the final dataset, which comprised of 3240 images, to further assess the model that demonstrated the best performance in the preceding phase For the initial phase the custom dataset was created from the 852 labeled images generated previously; of these, 792 images were used for training and the rest for testing. For the final phase 2592 images were used for training. Object detection is a computer vision task where the goal is to detect instances of objects belonging to a particular class within an image. It is a crucial step in many computer vision applications such as autonomous driving, surveillance and image retrieval (Vinogradova et al., 2002; Wang et al., 2018).

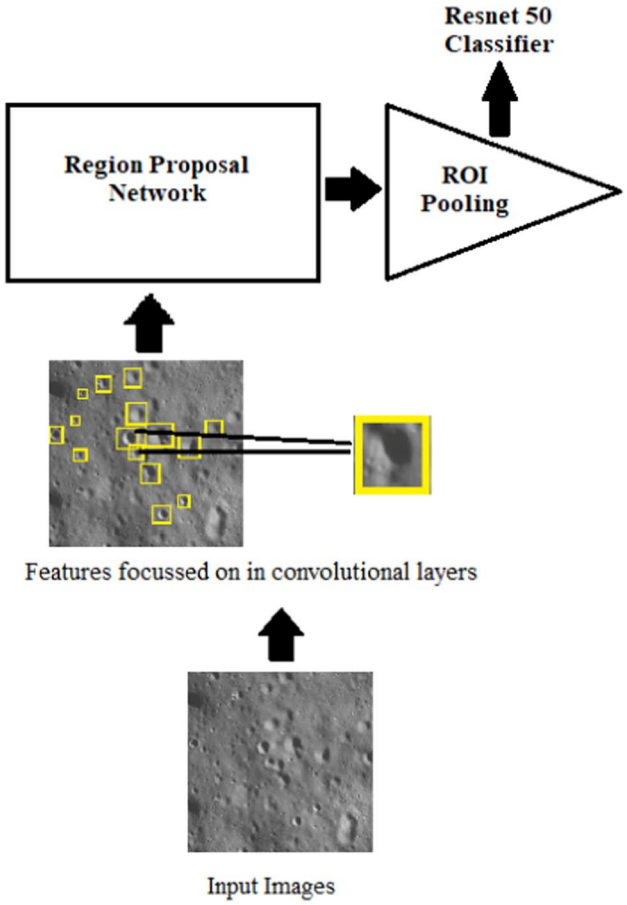


Fig. 4 Faster RCNN model flowchart (created)

Faster R-CNN, YoloV1, and YoloV5 are all state-of-the-art object detection models that have achieved impressive performance on benchmark datasets. However, each model has its own strengths and weaknesses. By training our custom dataset on all three models, the aim was to combine their strengths and achieve high accuracy and speed in object detection on our dataset.

In an optimal scenario for machine learning, there will be a large number of labeled training samples, but in reality this is not always the case. To address this issue transfer learning is used in machine learning which involves applying knowledge from one domain to another (Zhuang et al., 2021). We used transfer learning to fine-tune the pre-trained Faster R-CNN and YoloV5 models on the custom dataset. By using transfer learning, one can apply the knowledge gained by a pre-trained model on a large dataset to a custom dataset, thus leveraging existing knowledge for new tasks. This can significantly reduce the training time and improve the performance of the model on the dataset.

The backbone of the Faster R-CNN model is Resnet50 which is a neural network with 50 layers, which has its weights and biases trained on the COCO dataset He et al. (2015), and it is being fine-tuned to the custom dataset created by us. YoloV5m is also pre-trained on the COCO dataset. The YoloV5m network was trained twice, once using pre-trained weights for 200 epochs, and then the model was retrained from scratch on the custom dataset. Additionally, we trained custom YoloV1 and YoloV5 models from scratch on our dataset. Training from scratch allowed us to tailor the model to the specific dataset and potentially improve its performance.

Accuracy Assessment

In this framework, bounding boxes are used for image annotation. This annotator outlines the boundaries of the craters. To measure the correctness of the bounding box (bbox) predicted by the model with the bounding box of the label we used the Jaccard index. The Jaccard index is a measure of similarity between two sets of data. It is calculated as the ratio of the size of the intersection of the two sets to the size of their union Hancock (2004). Intersection of union (IoU) is the value given by the area of intersection between the label's bbox and model's bbox divided by the union of the label's bbox area and model's bbox area (Fig. 3).

If IoU value is equal to 1, then that implies that the bounding boxes are a perfect match. But taking the perfect match is not ideal. Therefore, thresholds are used. So, if the IoU value is greater than a threshold, it will be considered a true positive. We have used mean average precision (mAP) as the evaluation metric and considered various IoU values as thresholds to calculate the mAPs.

Proposed Framework

Faster R-CNN, YoloV5m, and YoloV1 are all popular architectures for object detection; these architectures have been designed to perform this task efficiently and accurately. This architecture is known for its high accuracy but can be slower than single-stage architectures like Yolo. Yolo is known for its speed; it is most commonly used for real-time object detection. Later versions of Yolo, such as the YoloV5m, use an improved backbone network and additional techniques, such as anchor boxes, to improve its performance.

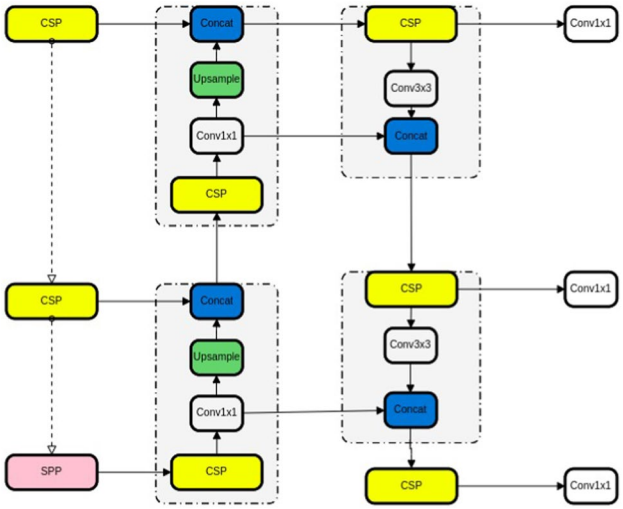


Fig. 5 YoloV5m model architecture

Faster RCNN

In this work, selective region-based R-CNN networks have been used for object detection. Each crater in the individual images was identified using bounding boxes. Faster R-CNN with Resnet50 as the backbone of the architecture was used as the model. The input to the model was a list of tensors for each corresponding image. During the training phase, the model accepted both input tensors and the target list of dictionaries containing: boxes containing craters and their respective labels for each image. The output in

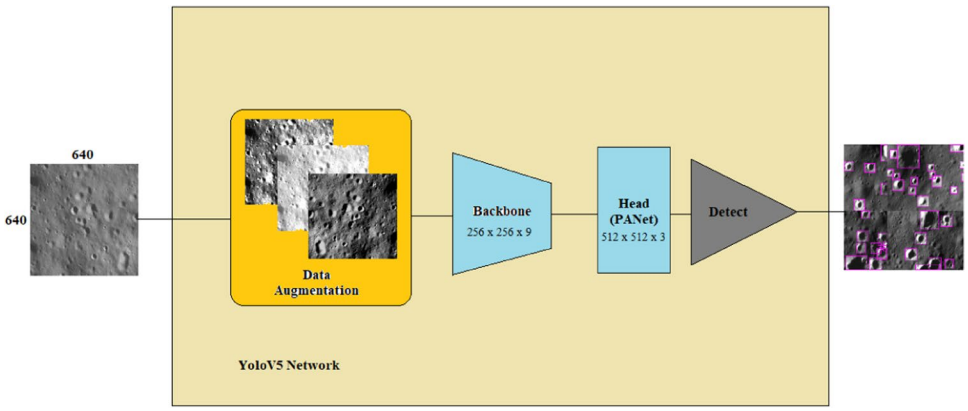


Fig. 6 YoloV5m process flow with custom crater images (Created)

Table 1 Yolov1 Architecture

Layer	Kernel size	Number of filters	Strides	Padding
Conv1	7	64	2	3
MaxPool1	2 × 2 Max Pooling			
Conv2	3	192	1	1
MaxPool2	2 × 2 Max Pooling			
Conv3	1	128	1	0
Conv4	3	256	1	1
Conv5	1	256	1	0
Conv6	3	512	1	1
MaxPool3	2 × 2 Max Pooling			
Conv7a	1	256	1	0
Conv7b	3	512	1	1
Conv7	Repeat 4 times: (Conv7a, Conv7b)			
Conv8	1	512	1	0
Conv9	3	1024	1	1
MaxPool4	2 × 2 Max Pooling			
Conv10a	1	512	1	0
Conv10b	3	1024	1	1
Conv10	Repeat 2 times: (Conv10a, Conv10b)			
Conv11	3	1024	2	1
Conv12	3	1024	1	1
Conv13	3	1024	1	1

Table 2 FasterRCNN results

Average precision (AP)	@ IoU = range(0.50, 0.95)	0.375
Average precision (AP)	@ IoU = 0.50	0.798
Average precision (AP)	@ IoU = 0.75	0.269

Table 3 Fine tuned YoloV5

Average precision (AP)	@ IoU = 0.50:0.95	0.382
Average precision (AP)	@ IoU = 0.50	0.791

the training phase was a dictionary of tensors that contained the classification and current regression loss that it faced, whereas, in the testing phase, the model could work with just the input tensor. Training this model was particularly tricky because of the varying sizes of craters within the same image. We were able to train the non-frozen layers on our dataset to obtain the final desired model that can detect craters.

Figure 4 depicts the process flow of the Faster R-CNN network. R-CNN approach to object detection looks at regions within the image and not at each image as a whole. Since this takes a lot of time as taking a huge number of regions as input to the CNN is heavy, we used an alternate version of this network known as Faster R-CNN. This method made the process many folds faster by passing the image into the convolutional layer rather than passing the region proposals Girshick (2015). The region proposal network (RPN) utilized the convolutional feature map to predict the proposal regions. The learned RPN improved the quality of the region proposal, thereby increasing the overall accuracy of the object detection model Ren et al. (2015). The predicted proposal regions were then warped into squares with the help of an ROI pooling layer and reshaped into a fixed size. It was then fed into a fully connected layer. The resulting ROI feature vector, coupled with a softmax layer, can predict the class of the proposed region, which in our case is just one(crater), and also the bounding box offset values.

YoloV5m

Another approach to detecting craters was making use of different YoloV5 networks (Zhu et al., 2021). YoloV5m is an object detection model that uses single-stage detector architecture. It is a variant of the YOLO (You Only Look Once) family of models, which means it takes an input image and outputs a set of bounding boxes and corresponding class

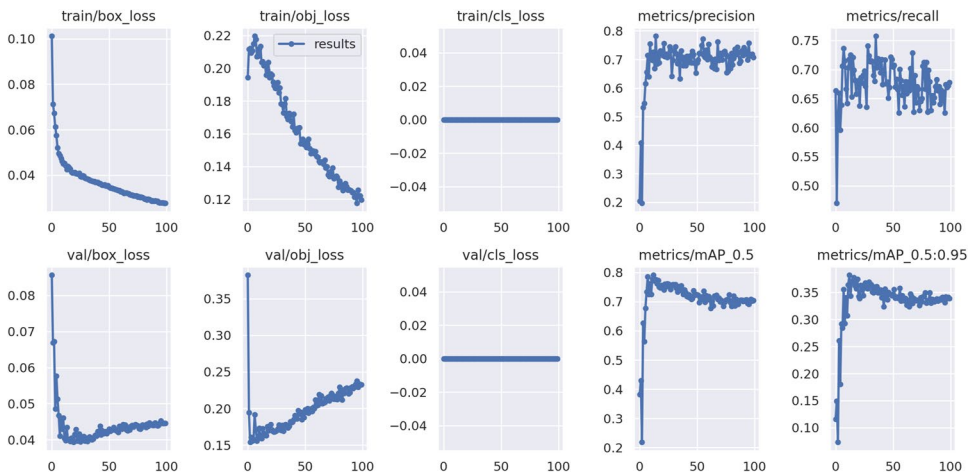


Fig. 7 YoloV5m pretrained results

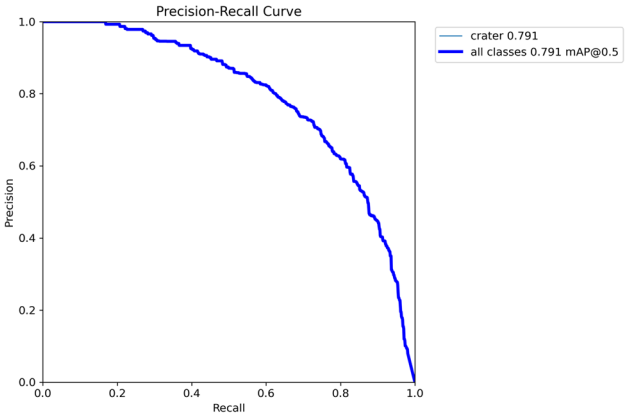


Fig. 8 Precision recall curve—YoloV5m pretrained

probabilities for the objects in the image (Karthi et al., 2021).

At a high level, YoloV5 uses a backbone network to extract features from the input image, followed by a neck network that performs feature selection and integration, and finally a head network that produces the object detection output.

The YoloV5m architecture consists of a backbone network, neck network, and head network. The backbone network is based on CSPNet, a type of convolutional neural network that uses cross-stage partial connections to improve performance. The neck networks is composed of SPP (Spatial Pyramid Pooling) and PAN (Path Aggregation Network) modules, which help to capture multi-scale features and integrate information from different levels of the feature hierarchy. The head network in YoloV5 then uses the selected features to predict the object detection output.

The head network consists of several prediction layers, each of which predicts bounding boxes and class probabilities for a different scale. By using multiple prediction layers, YoloV5 is able to accurately detect objects at different scales and in various contexts. This is how the YoloV5m network incorporates feature selection (Horvat et al., 2022).

Figure 5 shows that the YoloV5m model has a total of 18 convolutional layers in the backbone network, with channel counts ranging from 32 to 640. The neck network has two SPP modules and a PAN module, which allows the model to efficiently capture information at multiple scales. The head network has three prediction layers, each of which predicts bounding boxes and class probabilities for a different scale. This architecture has a total of 21 million parameters (DeLatte et al., 2019).

The process flow of an image in this network is shown in Fig. 6, where the image first goes through 3 augmentation processes, and the generated images are passed as input into the backbone network, followed by the PAN module and finally the output.

YoloV1

Since Yolo models performed really well on this dataset, we also tried implementing the YoloV1 model (Redmon et al., 2015) which is a very quick object identification method that analyses pictures at a rate of 45 frames per second in real time. The idea behind YOLO is relatively uncomplicated. For every bounding box, a single neural network predicts both the box itself and the probability of the object it contains.

The YOLOV1 algorithm concurrently predicts bounding boxes and incorporates characteristics from the full picture. By dividing the picture into a grid of S x S squares, the algorithm produces B bounding boxes and

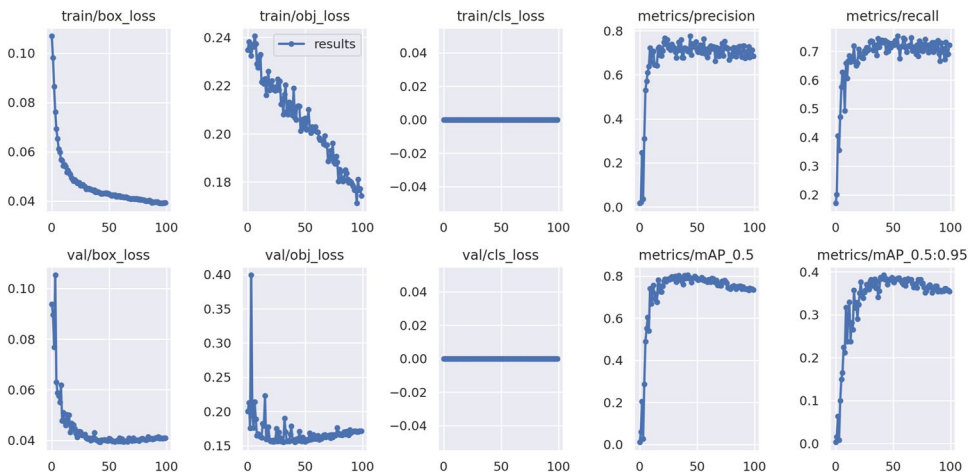


Fig. 9 YoloV5m trained from scratch results

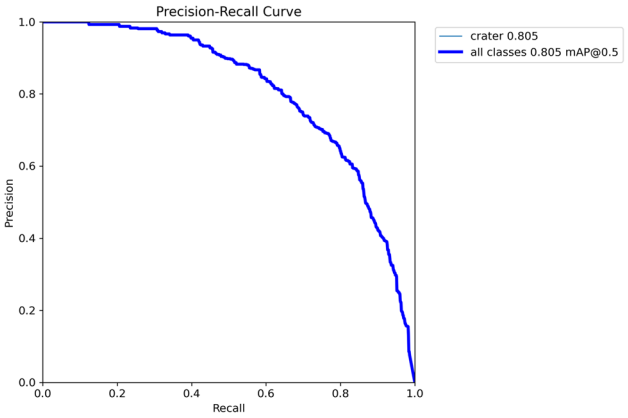


Fig. 10 Precision recall curve—YoloV5m trained from scratch

their respective confidence scores for each square. The model's confidence ratings provide insight into both its level of confidence that a given box contains an object and the accuracy of its prediction for that box.

This algorithm uses a combination of convolutional layers and fully connected layers to generate predictions. The convolutional layers extract features from the input image, while the fully connected layers make predictions about the output's probabilities and coordinates. The network includes 24 convolutional layers and 2 fully connected layers. The network layers are given in Table 1.

The main limitation faced while implementing this model was that Yolo's bounding box predictions are subject to substantial spatial restrictions as a result of each grid cell's inability to predict more than two boxes with a single class (Diwan et al., 2022). Spatial restrictions constrain the model's capacity to predict neighboring items.

Results and Discussion

Faster RCNN

Table 2 shows the IoU values and corresponding AP for the Faster RCNN network on the initial dataset. The Faster RCNN model is pretrained on the COCO dataset.

Table 4 YoloV5 from scratch

Average precision (AP)	@ IoU=0.50: 0.95	0.393
Average precision (AP)	@ IoU=0.50	0.805

Table 5 Custom Yolo V1

Average precision (AP)	@ IoU = 0.30	0.632
Average precision (AP)	@ IoU = 0.50	0.469

Yolo V5m

The fine-tuned YoloV5m model obtained had a decent Average Precision (AP) value for IoU set to 0.50. Table 3 shows the results obtained for the fine-tuned YoloV5m.

Over 200 epochs, the data collected on each epoch's performance, has been constructed into graphs and is represented in Fig. 7.

The precision-recall curve can be used to visualize the trade-off between precision and recall for different classification thresholds. A perfect model would achieve both maximum precision and recall, while a poor model would have low precision and recall values. From Fig. 8, it is a good sign that the model is fairly competent at identifying craters with a reasonably high precision until it reaches a recall of around 0.6.

While training YoloV5m from scratch, the model started out with an AP value as low as 0.003, and it ended up gaining the highest AP value. Table 4 shows the results of YoloV5m trained from scratch. The YoloV5m model was trained from scratch for over 200 epochs and the data collected has been constructed into graphs in Fig. 9. The precision-recall curve obtained from this model shown in Fig. 10 tells us that it has similar characteristics as the previous one.

Yolo V1

The YoloV1, which was trained from scratch, had the worst performance. This can be attributed to the inferior architecture of the YoloV1 model when compared to YoloV5 and faster RCNN. Even with a very low IoU threshold value of 0.30, it was only able to get an average precision of 63.2%. Table 5 depicts the performance of the custom YoloV1 trained from scratch.

Table 6 YoloV5 results on larger data

Average precision (AP)	@ IoU = 0.50: 0.95	0.55
Average precision (AP)	@ IoU = 0.50	0.90

Table 7 Precision, recall and F1 score for faster RCNN (larger data)

Precision	0.92
Recall	0.83
F1 Score	0.87

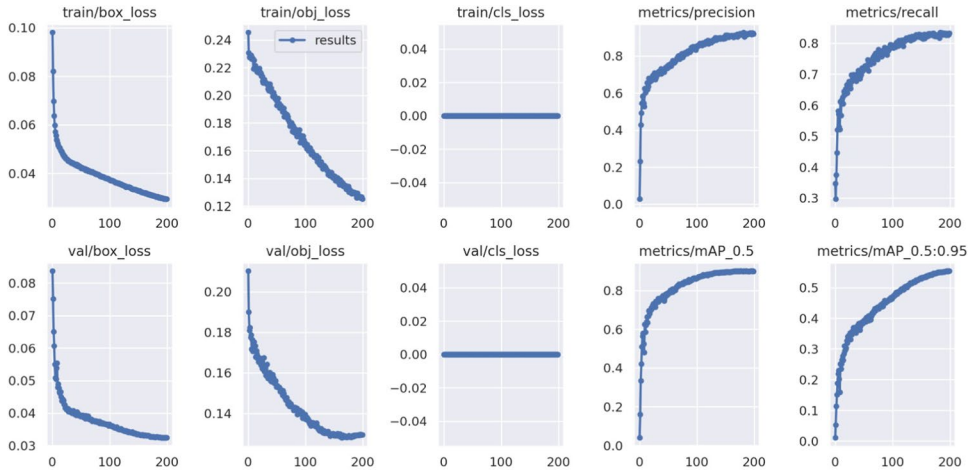


Fig. 11 YoloV5m trained using bigger data

Yolov5 on Larger Data

The Yolo V5m model was observed to have the best performance out of the three models when trained on the initial dataset containing 852 images. This model was then trained with the dataset with 3240 images.

The model's performance increased drastically by increasing the number of images used. Table 6 shows the Mean Average Precisons at the two IoU thresholds. The precision, recall and F1 scores are shown in Table 7. The true detection rate is the recall itself while the false detection of the model was 0.20. In the context of crater detection, the high recall of 0.83 is beneficial as it means the model is correctly identifying 83% of actual craters. This is crucial in ensuring that potential craters are not missed, which could be detrimental in planning missions or studying celestial bodies. However, the False Discovery Rate (FDR) of 0.20 indicates that 20% of the detections made by the model are false positives, meaning these are not actual craters but the model identified them as such. This could be due to human errors in manual labelling. The model detects the craters even if a particular crater was missed during labelling.

Analysis

In Figs. 7, 9 and 11 we compare the performance of three models for crater detection: a pretrained model with transfer learning on 852 images, a model trained from scratch with 852 images, and a model trained from scratch with a larger dataset of 3420 images. We use the following metrics to evaluate the models: losses (train/box_loss, train/obj_loss, val/box_loss, val/obj_loss, train/cls_loss, val/cls_loss), precision, recall, and mean average precision (mAP) at different intersection over union (IoU) thresholds.

The model trained from scratch with 852 images shows decreasing losses, indicating learning, but with some fluctuations in the validation losses which suggest some instability in learning from the validation set. The classification loss is relatively stable, which is expected in binary classification tasks if the model has learned to differentiate the classes well. Precision and recall are improving, but precision has a notable upward trend, indicating the model is getting better at making correct positive predictions over time. The mAP scores are increasing, showing the model's improved prediction accuracy over different IoU thresholds.

The pretrained model with transfer learning on 852 images shows lower and smoother losses compared to the model trained from scratch, suggesting that transfer learning provides a better starting point and more stable learning. Precision and recall are both high, with recall being notably stable and high throughout the training process, indicating that the model is reliably identifying most positives. The mAP scores are also high and stable, particularly the mAP at IoU 0.5, indicating good prediction accuracy.

The model trained with a larger dataset of 3240 images shows consistently low losses, especially the validation losses, which suggests that having a larger dataset contributes to better generalization and stability. Precision and recall both show an upward trend, with recall nearly reaching a plateau, which suggests that the model has a high true positive rate. The mAP scores improve significantly over the course of training, with the mAP at IoU 0.5 reaching high values, indicating excellent prediction accuracy. The mAP over a range of IoU thresholds (0.5:0.95) also reaches higher values than the previous models, showing that the model performs well even under stricter criteria.

When using transfer learning on faster RCNN, the average precision of IoU thresholds from 0.50 to 0.95 is 37%. The value obtained here is less than ideal; this is because as the IoU threshold increases, the bounding box has to get extremely close to the label's bounding box, so it need not be considered. The best results were obtained when the IoU threshold was set to 50%. A precision of 79.8% was obtained with that threshold. In general, the lower the threshold, the better the precision; hence, the 0.5 IoU threshold is a perfect

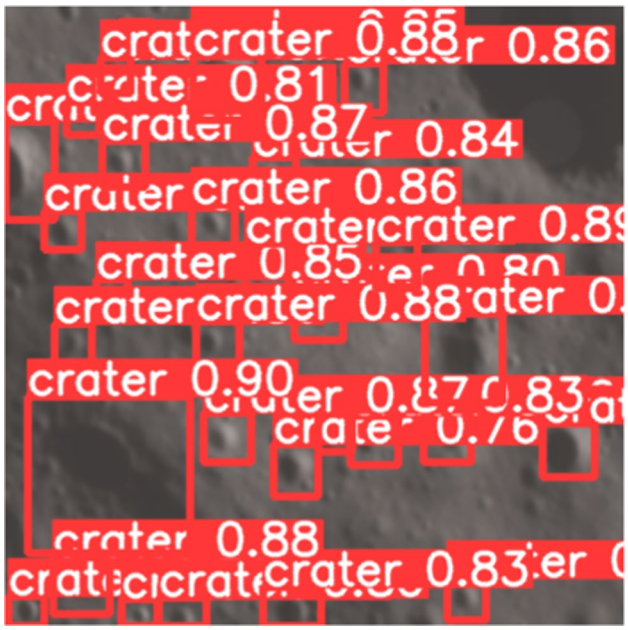


Fig. 12 Model labeled crater. Geospatial coordinates: Top-left (4.566241, 230.967869), Lower-right (4.571820, 230.962127)

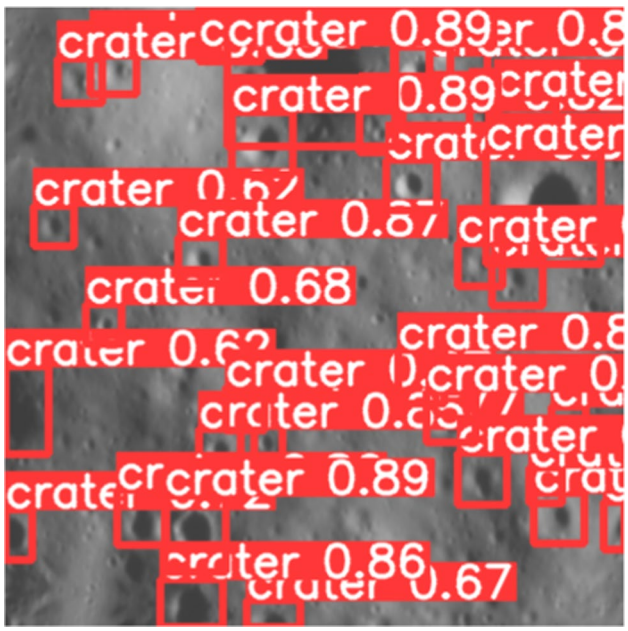


Fig. 13 Model labeled crater. Geospatial coordinates: Top-left (4.499287, 230.967869), Lower-right (4.504866, 230.962127)

balance. When it comes to YoloV5m, a slight increase in performance was observed by using transfer learning. It was able to get a mean average precision of 0.382, with IoU thresholds varying from 0.50 to 0.95. The best-performing model was also obtained using this architecture by training it from scratch on the custom dataset.The model was able to get an average precision of 80.5% with the IoU threshold set to 0.50. The performance of this model was further increased in phase 2 after it was trained on a larger dataset with 3240 images. The Yolo V5 model was able to get an average precision of 90% with the IoU threshold set to 0.50. A few examples to show the craters identified by the model (YoloV5) are shown in Figs. 12 and 13.

The bounding boxes currently lack geographical data; if it can be incorporated into the bounding boxes, then these bounding boxes can be used in Lunar Crater size-frequency distribution (SFD) models. These models are used to analyze the distribution of impact craters on the lunar surface. To use the bounding boxes for SFD, we would need to measure the size of each crater within its bounding box. This can be done by measuring the diameter of the crater, which is the longest distance across the crater. Once the size measurements for all the craters in a region are obtained, the SFD of a geographic region can then be plotted by counting the number of craters in each size range and plotting the results.

Conclusion

In this paper, we introduce a robust deep learning framework designed for the detection of lunar surface craters. The experimental evaluation incorporates Fast RCNN and Yolo models, revealing their enhanced resilience to image variations and distortions. The results highlight the efficacy of these deep learning models, particularly emphasizing the superior performance of the YoloV5m model. With an average precision (AP) of 0.382 at the intersection over union (IoU) ranging from 0.50 to 0.95, the YoloV5m model demonstrates remarkable adaptability to the diverse shapes, sizes, and environmental conditions of lunar craters. The lunar crater detection model, leveraging high-resolution imagery from the Chandrayaan-2 Orbiter, represents a significant advancement in astronomical feature analysis. The model exhibits an exceptional capability to efficiently process image data, segmenting images into 120 × 120-m tiles that retain a high level of detail. It is especially remarkable because it can identify craters with a diameter ranging from 120 m to only 10 m, indicating a degree of accuracy that exceeds that of many of its counterparts (Zhao & Ye, 2023). It is also able to detect partial craters as well, which can be observed in craters at the edges in Fig. 13. The proposed model exhibits a high degree of versatility, extending its applicability beyond OHRC images to include other types such as LRO NAC and TMC-2 images. Its ability to deliver accurate outputs across these varied image types underscores its generic nature. This study holds considerable promise for advancing our understanding of lunar geography

and assisting in the planning of future lunar exploration missions.

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Declarations

Conflict of interest The authors declared that they have no Conflict of interest.

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